

DXGAT / RampVision — technical audit: the current state

6 October 2026 · Valentyn Fedorov · an inventory of what runs today: the modules, the models they run and their versions, what each reads from the shared layer, in which window of the turnaround it is active, how long it runs, and where the logic repeats. Sources: the static architecture review of 7 September (docs/arch_review), the consumption matrix and the model audits (docs/analysis), and the measurements of 8 September – 1 October (post jobs on the 36-minute reference event zHxIAF2vUGxJ; batch medians per recorded frame on the contended test-set machine; live milliseconds per frame on an RTX 5070 Ti with the card kept busy). Generated by scripts/reports/build_audit_docx.py from docs/reports/audit_inventory.json.

1 · What runs today, in what order, and how long it takes

Everything is post-processing. The CameraBox records two 1080p streams at 8 fps (H.264, constant 4.0 Mbit/s each) in 1-minute chunks and uploads them to the bucket; after the turnaround the chunks are merged into one full-turn video per camera; then three GKE jobs run in sequence on that file — the General Model, the trackers, and one job per module — and the verdicts (Pass / Fail / Not observed) go to MongoDB and RampVision. On the median of 148 production videos (87 minutes of recording) the result arrives 20.3 h after the recording: 9.5 h for the last chunk to reach the bucket, 1.7 h waiting for merge, 0.8 h merging, 2.4 h of General Model (207 ms per frame, 1.7× slower than the recording), 1.3 h of trackers, 0.9 h of modules. A production GPU job gets a Tesla T4 (14.6 GB), 3.9 vCPU of a 2013-generation Xeon and 12 GB of RAM.

stage	repository · entry	what it computes	time on the median video (87 min)	per frame
merge	camera_software · merge.py	finds, orders and merges the chunks; capture-boundary classifier	1.7 h waiting + 0.8 h running	—
General Model	general_model · main.py (job per video, two decodes)	three YOLO heads, MobileSAM mask of the main aircraft, entity, camera type, aircraft type; the second run labels obstacle rows and the main aircraft	2.4 h	207 ms (prod T4); 19.2 ms busy on the RTX 5070 Ti, 11.1 with TensorRT
trackers	cv_trackers · tracker.py (job per video, pixels every frame)	three DeepSORT instances, instance masks, optical flow, state machines per object, noise estimate	1.3 h	112 ms (prod); 26.2 ms busy on the RTX 5070 Ti (4 classes)
modules	dxgat/detectors/<27 repos> · main.py (job per module)	the check logic on GM + tracker rows (and pixels for 13 of them)	0.9 h in total	0.1–18.8 ms of own work live; 1.2–30.7 ms per recorded frame in batch
reporting	db_worker · send_report.py	verdicts, reports, smart timeline → MongoDB, buckets, RampVision	minutes	—

2 · The shared layer: models that run for every module

General Model job (general_model, production commit a0157a4): fp16 ONNX heads through onnxruntime CUDA; the other models in torch. Milliseconds: busy card, RTX 5070 Ti, the head's share of the three-head step.

model	file (version)	input	when it runs	size MB	ms per frame
GM YOLOv8m (apron classes)	GM_yolov8m_best_augmentation_march2024.onnx	1088x1088 fp16	every frame	6.2	8.5
Chocks YOLOv8 v4.3	chocks_v4.3_200ep_yolov8.onnx	1280x1280 fp16	every frame	22.6	3.0
Vehicle YOLOv8m	VM_yolov8m_last_september2023.onnx	1088x1088 fp16	every frame	52.0	5.8
MobileSAM vit_t (main-aircraft mask)	mobile_sam.pt	1024 (full frame)	every frame with the main aircraft	40.7	—
Entity (airline) YOLOv5	entity_det_aug_expnd.pt	1280	every 8th frame, up to 40 hits	14.7	—
Camera cone/wing EfficientNet-B0 v1.8.1	camera_cls_effnet_b0_october_v1.8.1.pt	224 (rows 150:930)	every post-arrival frame with the main aircraft, CPU	16.3	—
Tail classifier	general_tail_classifier.pth	crop	aircraft type vote	0.4	—

Trackers job (cv_trackers, production commit bd43c3c with cv_common tracker_optimization @2759daf): one tracker stream per video, four classes (aircraft, belt loader, GSE, person); the production job costs 36–49 ms per frame on this card, the v2 port is byte-identical.

component	file	input	when it runs	size MB	ms per frame (busy, thread time)
DeepSORT re-id, workers (TorchScript)	scripted_ckpt.t7	64x128 crops	every frame with persons	46.1	4.1

component	file	input	when it runs	size MB	ms per frame (busy, thread time)
DeepSORT re-id, belt loaders and GSE (torchvision ResNet-34, ImageNet, training mode)	torchvision resnet34	128x64 crops	every frame with BL / GSE, two instances	87.3	15.2
Plane segmentor YOLO11s-seg	yolo11s-seg_plane.pt	640	at keypoint (re)initialisation, every 8 frames per object	20.6	—
BL / GSE segmentor YOLO26s-seg	yolo26s_seg_bl_gse_tr10_noalb.pt	640	the same, cached per class per frame	23.4	4.2
FAST keypoints + PyrLK optical flow (OpenCV, CPU)	—	<=250 points per object	every frame	—	12.0
estimate_sigma noise estimate (scikit-image, CPU)	—	full frame	every 2 s	—	2.1

3 · The modules: what each one checks, when it is active, what it reads, what it runs

27 module repositories implement the client's 30 checks (aircraft-chocks gives three verdicts; “proper belt loader approach” is derived from 3-stop and hand signals). Window: the stage the module is gated to and the events that open and close it (from the streaming audit of the code); decision: INSTANT = one detection → Fail, EVENT = verdict at the event, DEADLINE = Fail if not seen by T, LOOKBACK = the window before T, WHOLE = the whole event. GM classes: what the module reads for its decision (drawn-only classes excluded). Pixels: whether it decodes frames. Passes: how many times it reads the whole recording.

M	module (repository)	check	tier · cameras	window · decision	GM classes read	tracker classes	passes · pixels	own models
M01	3-stop-brake-check	3 stop brake check	streaming · Aircraft=Both, Jet=Wing	DOWNLOAD: beltloader appeared → BL stopped next to the aircraft (... · EVENT	gse, pushback, trailer	airplane, beltloader	1 · no	none
M02A	aircraft-chocks	Nose gear chocks applied immediately; Main gear chocks remo...	real-time · Aircraft=Cone, Jet=Cone	PRE_DEP: T_arr → pushback attached → chock removal · EVENT	pushback, chock, cone, trailer, fuel_truck, ladder, front_w...	airplane, person, gse, beltloader	2 · yes	frame-difference chock classifier (rear...
M03	all-cargo-bin-doors-opened-and-verified	All cargo bin doors opened and verified	post · Aircraft=Both, Jet=Wing	DOWNLOAD: T_arr → end of the stage or 480 frames of... · DEADLINE	front_door, back_door, gse, pushback, trailer, fuel_truck, ...	airplane	1 · no	none
M04	beltloader-chocks	Belt loader forward chock remained in place until unit is b...	real-time · Aircraft=Both, Jet=Wing	DOWNLOAD: BL next to the aircraft → BL drove away · EVENT	chock, cone, back_wheel, airplane_engine, back_door, airpla...	airplane, beltloader	1 · no	none
M05	bl_rear_cone	Beltloader rear cone positioned after BL is in place	real-time · Aircraft=Both, Jet=Wing	DOWNLOAD: BL next to the aircraft → BL confirmed in place · EVENT	beltloader_cone, engine_cone, wing_cone, cone, back_wheel, ...	airplane, beltloader	1 · no	none
M06	chocks-and-cones-available-and-staged-for...	Chocks and cones available and staged for arrival	post · Aircraft=Cone, Jet=Cone	PRE_ARR: start of the stream → T_arr · LOOKBACK	obstacle, side_obstacle, beltloader, gse, pushback, cone, c...	airplane	1 · no	none
M07	conditioned-air-removed-10-mins-prior-to-...	Conditioned air removed 10 mins prior to departure and prop...	streaming · Aircraft=Cone, Jet=Cone	PRE_DEP: T_dep – 10 min (in practice: BL l... → T_dep · LOOKBACK	trailer, gse, beltloader, front_wheel, airplane_nose, pushb...	airplane	1 · yes	AC hose detector (YOLO11s-seg, 1 class)
M08	cones-are-removed-only-after-all-gse-is-c...	Cones removed only after all GSE is clear and chocked	real-time · Aircraft=Cone, Jet=Wing	PRE_DEP: BL leave → T_dep · EVENT	engine_cone, wing_cone, tail_cone, gse, pushback, trailer, ...	airplane, beltloader	1 · no	none
M09	cones-placed-in-proper-positions-and-time...	Cones placed in proper positions and timely	real-time · Aircraft=Both, Jet=Both	ARR_POST: T_arr → T_arr + time limit · DEADLINE	wing_cone, engine_cone, tail_cone, airplane_tail, gse, push...	airplane	1 · no	none
M10	crew-present-10-minutes-prior-to-aircraft...	Crew present 10 minutes prior to aircraft arrival	post · Aircraft=Cone, Jet=Cone	PRE_ARR: start of the stream → T_arr · LOOKBACK	person, gse, pushback, trailer	airplane	1 · no	none
M11	fod-walk-completed	FOD walk completed	streaming · Aircraft=Cone, Jet=Cone	PRE_ARR: start of the stream → T_arr · LOOKBACK	front_wheel, gse, beltloader, pushback, airplane_tail, airp...	airplane, person	1 · yes	MoveNet SinglePose Thunder v4
M12	gse-chocks	Motorized GSE parked and properly chocked	streaming · Aircraft=Both, Jet=Wing	DOWNLOAD: GSE stopped next to the aircraft → GSE drove away · EVENT	chock, pushback, trailer, fuel_truck, cone, ladder	gse, person, beltloader	1 · yes	frame-difference chock classifier; RTMPose-l body pose (17 keypoints)

M	module (repository)	check	tier · cameras	window · decision	GM classes read	tracker classes	passes · pixels	own models
M13	hand-signals	All GSE guided into aircraft using approved hand signals	post · Aircraft=Both, Jet=Wing	ARR_POST: T_arr → GSE pulled up to the door · EVENT	front_door, back_door, pushback, back_wheel, airplane_engin...	airplane, beltloader, person	2 · yes	HRNet-W48 top-down heatmap pose (COCO-1...
M14	handrails-on-gse-being-used	Handrails on GSE being used	streaming · Aircraft=Both, Jet=-	DOWNLOAD: BL next to the aircraft → worker got off the BL · EVENT	person, trailer, gse, airplane_engine, back_wheel, back_door	airplane, person, beltloader	1 · yes	EfficientNetB0 handrail-extension class...; SegFormer-style handrail segmentation (...; DWPose-l whole-body pose (133 keypoints...; Hand-contact classifier (blind checks)
M15	lead-marshaller-and-wing-walkers-in-posit...	Lead marshaller and wing walkers in correct position	real-time · Aircraft=Cone, Jet=Cone	PRE_ARR: start of the stream → T_arr · LOOKBACK	airplane_tail, airplane_wing, beltloader, gse, pushback, tr...	airplane, person	1 · no	HRNet-W48 top-down heatmap pose (COCO-1...; worker role classifier (SVC); walk-along classifier (RandomForest); walker movement-start models (left, rig...
M16	pin-verification	Pushback operator verifies steering bypass pin installation	streaming · Aircraft=Cone, Jet=-	PRE_DEP: BL leave → pushback attached · EVENT	pushback, airplane_nose, front_wheel, beltloader, gse, trai...	airplane, person	2 · yes	SimCC ResNet-50 top-down pose (COCO-17)
M17	post-arrival-aircraft-walk-around-inspect...	Post-arrival walk around inspection completed accurately; (...)	post · Aircraft=Cone, Jet=-	ARR_POST: T_arr → end of the walk-around · WHOLE	beltloader, fuel_truck, gse, ladder, trailer, back_wheel, a...	airplane, person, beltloader	1 · yes	YOLOv8m-pose (worker keypoints; singlet...; Depth Anything V2 metric ViT-B; MobileSAM vit_t (TDV masks); Walk-around canvas CNN (TorchPredictor ...)
M18	pre-arrival-safety-huddle	Safety huddle conducted at huddle cone	post · Aircraft=Cone, Jet=Cone	PRE_ARR: start of the stream → T_arr · LOOKBACK	cone, beltloader, gse, pushback, trailer, fuel_truck	airplane, person	1 · yes	none
M19	pre-departure-walk-around-completed	Pre-departure walk around completed	post · Aircraft=Cone, Jet=-	PRE_DEP: BL leave → T_dep · WHOLE	beltloader, gse, pushback, trailer, airplane_engine, cone, ...	airplane, person, beltloader	1 · yes	YOLOv8m-pose (worker keypoints, singlet...; YOLOv8m-pose rescue re-track (second YO...; Depth Anything V2 metric ViT-B; MobileSAM vit_t (TDV masks); Walk-around canvas CNN best_acc.keras v...
M20	pushback-does-not-start-until-wing-walker...	Pushback does not start until wing walkers are in place and...	streaming · Aircraft=Cone, Jet=Cone	DEP: pushback attached → pushback started moving · EVENT	airplane_tail, airplane_wing, pushback, front_wheel, beltlo...	airplane, person	1 · no	none
M21	pushback-pathway-confirmed-clear-of-obsta...	Pushback pathway confirmed clear of obstacles	real-time · Aircraft=Cone, Jet=Cone	DEP: T_dep – 45 s → T_dep · INSTANT	pushback, airplane_tail, airplane_nose, beltloader, gse, fu...	airplane	1 · no	none
M22	safety-handrails-fully-extended	Safety handrails fully extended and used	streaming · Aircraft=Both, Jet=-	DOWNLOAD: BL next to the aircraft → BL drove away · EVENT	person, trailer, gse, back_wheel	airplane, beltloader	1 · yes	EfficientNetB0 handrail-extension class...
M23	safety-vests-secured-to-body	Employees wearing safety vests secured to body	streaming · Aircraft=Both, Jet=Both	ALL: start of the stream → end of the stream · INSTANT	safety_vest	person	1 · yes	Swin-T orientation classifier (back / h...; Swin-T vest classifier (unzipped / zipp...; ResNet-18 vest-presence gate; YOLOv8m-pose (v4 belly-visible gate); ResNet-18 small-band vest CNN (v4_cnn2)
M24	safety-zone-confirmed-clear	Safety zone confirmed clear	streaming · Aircraft=Cone, Jet=Cone	PRE_ARR: T_arr – 2 min → T_arr · LOOKBACK	cone, front_wheel, person, pushback, beltloader, gse, fuel_...	airplane	2 · yes	MobileSAM vit_t (SamPredictor, box prom...
M26	steering-by-pass-pin-installed-or-steerin...	Steering by-pass pin installed, or steering otherwise bypas...	streaming · Aircraft=Cone, Jet=-	ARR_POST: T_arr → T_arr + 240 s · DEADLINE	person, front_wheel, chock	airplane, person	1 · yes	HRNet-W48 top-down heatmap pose (COCO-1...; tsai InceptionTime pin-installation cla...; MinMaxScaler
M27	wing-walkers-in-proper-position-and-using...	Wing walkers in proper position and using approved wands	streaming · Aircraft=Cone, Jet=Cone	DEP: T_dep – predeparture_prepare → T_dep + 60 s · LOOKBACK	airplane_engine, airplane_wing, airplane_nose, airplane_tai...	airplane, person	1 · no	HRNet-W48 top-down heatmap pose (COCO-1...

4 • Versions, pins and run times

Reviewed pin: the commit the architecture review inspected (7 September); head: the checkout the monthly test set runs. cv_common / db_worker: the submodule revisions the module is pinned to (ten different cv_common pins across the set). Post job: a clean single job on the 36-minute reference event. Batch: median ms per recorded frame on the contended test-set machine (model loading and decoding included; an early-exiting module looks cheaper). Live: the module's own work per frame inside the real-time branch, the shared GM and tracker excluded. Real time as is: whether the unchanged module runs live with the post-processing verdict.

M	module	reviewed pin	head	cv_common	db_worker	post job s	batch ms	live ms	real time as is	verdict arrives (live)
M01	3-stop-brake-check	3a2337c1	14685f4 (≠ pin)	86731e4c	5a4aa83d	16	3.4	0.64	yes	24:25 (T_dep + 0 s)
M02A	aircraft-chocks	fa00478f	fa00478	ac5098d2	5a4aa83d	38	5.7	0.48	no: two passes	—
M03	all-cargo-bin-doors-opened-and-verified	ac2707cb	c60aed2 (≠ pin)	dd5b5547	6efc0b05	8	1.4	0.87	yes	8:44 (T_arr + 164 s)
M04	beltloader-chocks	7056099b	7056099	ac5098d2	5a4aa83d	12	2.7	0.34	yes	24:25 (T_dep + 0 s)
M05	bl_rear_cone	d75f1f8b	12429f5 (≠ pin)	86731e4c	5a4aa83d	12	—	0.37	yes	24:25 (T_dep + 0 s)
M06	chocks-and-cones-available-and-staged-for...	1413225f	1413225	dd5b5547	5a4aa83d	7	1.4	0.09	yes	6:00 (T_arr + 0 s)
M07	conditioned-air-removed-10-mins-prior-to-...	940fdcf5	940fdcf	dd5b5547	5a4aa83d	27	2.4	1.26	yes	24:35 (T_dep + 10 s)
M08	cones-are-removed-only-after-all-gse-is-c...	e2a25ba3	e2a25ba	33c188d0	5a4aa83d	20	4.9	1.54	yes	24:25 (T_dep + 0 s)
M09	cones-placed-in-proper-positions-and-time...	b2c53f69	b2c53f6	dd5b5547	5a4aa83d	13	2.3	0.62	yes, report shifts	24:35 (T_dep + 10 s)
M10	crew-present-10-minutes-prior-to-aircraft...	5cfd25c9	5cfd25c	dd5b5547	6efc0b05	12	1.2	0.54	yes	7:04 (T_arr + 64 s)
M11	fod-walk-completed	b923b5c8	b923b5c	ac5098d2	5a4aa83d	21	3.0	1.32	yes	6:00 (T_arr + 0 s)
M12	gse-chocks	3160b871	3160b87	dd5b5547	5a4aa83d	76	6.1	3.25	yes, verdict at session end	session end
M13	hand-signals	6bd027f2	60c0faf (≠ pin)	8e3a7bf6	5a4aa83d	91	10.1	0.28	no: two passes	—
M14	handrails-on-gse-being-used	59fbced4	59fbced	33c188d0	5a4aa83d	249	30.7	12.46	yes, verdict at session end	session end
M15	lead-marshaller-and-wing-walkers-in-posi...	9b37ccff	9b37ccf	dd5b5547	5a4aa83d	59	5.7	2.50	yes	6:10 (T_arr + 10 s)
M16	pin-verification	8e1863c7	8e1863c	dd5b5547	6efc0b05	27	3.9	0.44	no: two passes	—
M17	post-arrival-aircraft-walk-around-inspect...	a0d9ee4f	7112c99 (≠ pin)	bb0b1746	0248146c	own env	6.8	—	not hosted	—
M18	pre-arrival-safety-huddle	5c8a7599	9b61d2e (≠ pin)	acfc0808	6efc0b05	22	1.9	0.87	yes	7:00 (T_arr + 60 s)
M19	pre-departure-walk-around-completed	e407de23	f2b6c36 (≠ pin)	d7f907cd	0248146c	own env	26.1	—	not hosted	—
M20	pushback-does-not-start-until-wing-walker...	f3b041cb	f3b041c	dd5b5547	5a4aa83d	13	2.4	0.33	yes	24:39 (T_dep + 14 s)
M21	pushback-pathway-confirmed-clear-of-obsta...	104db85a	104db85	dd5b5547	5a4aa83d	16	3.4	0.48	yes, report shifts	24:46 (T_dep + 20 s)
M22	safety-handrails-fully-extended	8b9a2a9b	8b9a2a9	33c188d0	5a4aa83d	46	8.4	2.37	yes	24:25 (T_dep + 0 s)
M23	safety-vests-secured-to-body	ae8ef164	ae8ef16	4291b0d9	0248146c	370	28.2	18.75	yes, verdict at session end	session end
M24	safety-zone-confirmed-clear	91676a55	91676a5	dd5b5547	5a4aa83d	83	10.3	0.25	no: two passes	—
M26	steering-by-pass-pin-installed-or-steerin...	60dca736	60dca73	8e3a7bf6	5a4aa83d	26	1.9	0.73	yes	10:05 (T_arr + 245 s)
M27	wing-walkers-in-proper-position-and-using...	4800b649	5f7ca1a (≠ pin)	dd5b5547	5a4aa83d	111	3.2	5.14	yes	25:09 (T_dep + 44 s)

The heaviest by post-job time: safety 370 s, handrails-on-gse-being-u 249 s, wing-walkers-in-proper-p 111 s, hand-signals 91 s, safety 83 s, gse-chocks 76 s. Per stage in batch the peak is download/upload (belt loader at the door → leaves): handrails-on-gse 30.7 + vests 28.2 + safety-handrails 8.4 + gse-chocks 6.1 + aircraft-chocks 5.7 + 3-stop 3.4 + beltloader-chocks 2.7 + bl_rear_cone 2.4 + cargo 1.4 = 89 ms per recorded frame against a 125 ms budget; the walk-arounds (26.1 and 6.8) run in their own Python environments.

5 · Models inside the modules

From the read-only audits of the checkouts the test set runs (sha256 of every weights file; nothing executed). The shared GM and tracker models are in section 2.

M	module	model	architecture	weights file	MB	sha256	framework	when it is called
M02A	aircraft-chocks	frame-difference chock classifier (rear wheel...	timm efficientnet_b0 pretrained=False num_classes=1 (state dict, 360 ...	effnetb0_ext_fix_993.pt	—	8211b62aa9c367aa	torch + timm; OpenCV/PIL/torc...	["main.py:641 (initial checks)", "main.py:731 (main check)"] ...
M07	conditioned-air-removed-10-mi...	AC hose detector (YOLO11s-seg, 1 class)	ultralytics SegmentationModel yolo11s-seg, nc 1, task segment (checkp...	yolo11s_seg-det_ac_tr18_noalb.pt	—	1a24fb38ea03ae4f	ultralytics YOLO on torch eag...	["main.py:579-583 (inside get_ac_hose_detections, called fr...
M11	fod-walk-completed	MoveNet SinglePose Thunder v4	MoveNet SinglePose Thunder, ONNX from tf2onnx, opset 11, 229 nodes; i...	movenet_singlepose_thunder_4.onnx	25	7fae4dc3cdd07ebd	onnxruntime	—
M12	gse-chocks	frame-difference chock classifier	timm efficientnet_b0 pretrained=False num_classes=1 (state dict, 360 ...	effnetb0_small_clean_64_960.pt	—	e6453d73c4e59538	torch + timm; OpenCV/PIL/torc...	["main.py:1315-1318 (stop check)", "main.py:1451-1454 (depa...
M12	gse-chocks	RTMPose-l body pose (17 keypoints)	RTMPose-l SimCC head, head.final_layer [17,1024,7,7]; ONNX opset 11, ...	rtmpose-l-4dba18fc.onnx	—	cff059fd58a2c0d5	onnxruntime-gpu (providers CU...	["main.py:747", "onnx_model_utils.py:9-21, 128-150"]
M13	hand-signals	HRNet-W48 top-down heatmap pose (COCO-17)	mmpose TopdownPoseEstimator: HRNet-W48 backbone + HeatmapHead (17x48x...	td-hm_hrnet-w48_8xb32-210e_coco-256x192-Oe6...	269	0e67c6167d6a10fe	torch via mmpose 1.3.1 / mmen...	—
M14	handrails-on-gse-being-used	EfficientNetB0 handrail-extension classifier	torchvision efficientnet_b0(weights=None), classifier[1] = Linear(128...	efficientNetb0_300_epoch_general_classifier...	—	31aae690e256c54b	torch / torchvision	["main.py:1065", "local_models/efficientnet.py:39-48"]
M14	handrails-on-gse-being-used	SegFormer-style handrail segmentation (smp Un...	smp Unet, encoder mit_b0, 1-class head; 5 551 483 params, float32	safety_handrails_mit_b0_best_model_293_epoc...	—	13c5f321659ee6f3	torch + segmentation_models_p...	["main.py:1074", "local_models/mix_vision_transformer.py:40...
M14	handrails-on-gse-being-used	DWPose-l whole-body pose (133 keypoints incl....	RTMPose/SimCC head, head.final_layer [133,1024,7,7]; ONNX opset 11, i...	dw-ll_ucoco-384.onnx	—	724f4ff2439ed61a	onnxruntime (providers [CUDA...	["main.py:1212", "main.py:423-427", "local_models/dwpose/wh...
M14	handrails-on-gse-being-used	Hand-contact classifier (blind checks)	timm efficientnet_b0 (num_classes=0, created with pretrained=True) + ...	effnetb0_weighted_norm_aug_929.pt	—	7fc4e4fdf8db0d4d	torch + timm	["main.py:636", "main.py:673", "local_models/efficientnet_r...
M15	lead-marshaller-and-wing-walk...	HRNet-W48 top-down heatmap pose (COCO-17)	mmpose TopdownPoseEstimator HRNet-W48 + HeatmapHead (same checkpoint ...	td-hm_hrnet-w48_8xb32-210e_coco-256x192-Oe6...	269	0e67c6167d6a10fe	torch via mmpose 1.3.1	—
M15	lead-marshaller-and-wing-walk...	worker role classifier (SVC)	imblearn Pipeline(StandardScaler, SMOTE, SVC), scikit-learn 1.6.1 (pi...	worker_classifier_svc.pkl	—	6a049e293cf13f2d	scikit-learn 1.6.1 + imbalanc...	—
M15	lead-marshaller-and-wing-walk...	walk-along classifier (RandomForest)	imblearn Pipeline(SMOTE, RandomForestClassifier), scikit-learn 1.6.1	walk_along_classifier_rfc.pkl	2	74fb67d1012e8a19	scikit-learn / imbalanced-lea...	—
M15	lead-marshaller-and-wing-walk...	walker movement-start models (left, right)	sklearn Pipeline(StandardScaler, GradientBoostingClassifier), scikit-...	right_walk_start_prediction_sgb.pkl	4	["893f608cd177a307", "Secf5ddccf278a5d"]	scikit-learn	—
M16	pin-verification	SimCC ResNet-50 top-down pose (COCO-17)	mmpose TopdownPoseEstimator: ResNet-50 backbone + SimCCHead (deconv h...	simcc_res50_8xb32-140e_coco-384x288-45c3ba3...	178	9e3c7ee446c0cc3a	torch via mmpose 1.3.1	—
M17	post-arrival-aircraft-walk-ar...	YOLOv8m-pose (worker keypoints; singleton sha...	YOLOv8m-pose, 17 keypoints; checkpoint written by ultralytics 8.0.77,...	yolov8m-pose.pt	—	2d0c64408d6b4015	ultralytics 8.2.48 (venv)	["code/walk_around/detection_processor.py:53 (batched perso...
M17	post-arrival-aircraft-walk-ar...	Depth Anything V2 metric ViT-B	DINOv2 ViT-B/14 + DPT head, max_depth 80; 97 470 785 params	depth_anything_v2_metric_vkitti_vitb.pth	—	4dad67a7cc10b462	torch (vendored depth_anythin...	["code/walk_around/temp/plane_tdv_extractor.py:336 (TDV fra...
M17	post-arrival-aircraft-walk-ar...	MobileSAM vit_t (TDV masks)	TinyViT encoder + SAM decoder	mobile_sam.pt	—	6dbb90523a35330f	torch, mobile_sam	["code/walk_around/temp/plane_tdv_extractor.py:44-51 Object...
M17	post-arrival-aircraft-walk-ar...	Walk-around canvas CNN (TorchPredictor CNNMod...	small conv net (features 8->...) + MLP classifier ending Linear(128,1...	best_model.pth	—	480c6f871f125e8c	torch	["code/walk_around/walk_around.py:341", "code/walk_around/m...
M19	pre-departure-walk-around-com...	YOLOv8m-pose (worker keypoints, singleton)	YOLOv8m-pose, 17 keypoints	yolov8m-pose.pt	—	2d0c64408d6b4015	ultralytics 8.2.48 (venv)	["code/walk_around/detection_processor.py:44 (per person cr...
M19	pre-departure-walk-around-com...	YOLOv8m-pose rescue re-track (second YOLO ins...	YOLOv8m-pose boxes + BoT-SORT ids	yolov8m-pose.pt	—	2d0c64408d6b4015	ultralytics 8.2.48 track()	["code/walk_around/gs1737_addons.py:219-220"]
M19	pre-departure-walk-	Depth Anything V2 metric ViT-B	DINOv2 ViT-B/14 + DPT head	depth_anything_v2_metric_vkitti_vitb.pth	—	4dad67a7cc10b462	torch (vendored	["code/walk_around/temp/plane_tdv

M	module	model	architecture	weights file	MB	sha256	framework	when it is called
	around-com...						depth_anythin...	_extractor.py:371 (TDV fra...
M19	pre-departure-walk-around-com...	MobileSAM vit_t (TDV masks)	TinyViT encoder + SAM decoder	mobile_sam.pt	—	6dbb90523a35330f	torch, mobile_sam	["code/walk_around/temp/plane_tdv_extractor.py:45-52 (set_i...
M19	pre-departure-walk-around-com...	Walk-around canvas CNN best_acc.keras via tor...	VGG-style 10 conv + BN + 5 max-pool + 4 dense, sigmoid; input 224x224...	best_acc.keras	—	9b2df8cb8edd8bfe	PyTorch twin of a Keras 2.15 ...	["code/walk_around/walk_around.py:291 (per canvas)", "code/...
M22	safety-handrails-fully-extend...	EfficientNetB0 handrail-extension classifier	torchvision efficientnet_b0(weights=None), classifier[1] = Linear(128...	efficientNetb0_300_epoch_general_classifier...	—	31aae690e256c54b	torch / torchvision	["main.py:284", "efficientnet.py:39-48"]
M23	safety-vests-secured-to-body	Swin-T orientation classifier (back / head / ...	torchvision swin_t(num_classes=3) state dict (head 3x768)	pose_classifier.pt	110	175e87d3668a5c9c	torch / torchvision	—
M23	safety-vests-secured-to-body	Swin-T vest classifier (unzipped / zipped)	torchvision swin_t(num_classes=2) state dict (head 2x768)	vest_classifier.pt	110	b0a5f34b168a97f6	torch / torchvision	—
M23	safety-vests-secured-to-body	ResNet-18 vest-presence gate	torchvision resnet18(num_classes=2) state dict	presence_bbox_r18.pt	45	4bcb3e3f2a242f24	torch / torchvision	—
M23	safety-vests-secured-to-body	YOLOv8m-pose (v4 belly-visible gate)	ultralytics PoseModel (C2f, SPPF, Pose head) (pickle class scan)	yolov8m-pose.pt	53	dbe539ea268db253	ultralytics 8.4.83 on torch	—
M23	safety-vests-secured-to-body	ResNet-18 small-band vest CNN (v4_cnn2)	torchvision resnet18, fc -> 1 logit (v4_override.py:154-158)	v4_cnn2.pt	45	66d4a8f79a4843c2	torch / torchvision	—
M24	safety-zone-confirmed-clear	MobileSAM vit_t (SamPredictor, box prompt)	TinyViT image encoder (1024 input -> 256x64x64 embedding) + SAM promp...	mobile_sam.pt	—	6dbb90523a35330f	torch, package mobile_sam	["main.py:1426 -> cv_common/tracked_object.py:393-403 -> 38...
M26	steering-by-pass-pin-installe...	HRNet-W48 top-down heatmap pose (COCO-17)	mmpose TopdownPoseEstimator HRNet-W48 + HeatmapHead, MSRAHeatmap 48x6...	td-hm_hrnet-w48_8xb32-210e_coco-256x192-0e6...	269	0e67c6167d6a10fe	torch via mmpose 1.3.1	—
M26	steering-by-pass-pin-installe...	tsai InceptionTime pin-installation classifier	fastai Learner wrapping tsai.models.InceptionTime (Conv1d, BatchNorm1...	model_newaug.pkl	2	bfa75c6ce2f6e6da	fastai 2.7.18 + tsai 0.3.9 on...	—
M26	steering-by-pass-pin-installe...	MinMaxScaler	sklearn.preprocessing.MinMaxScaler, pickled with scikit-learn 1.6.1 (...)	scaler.gz	—	03248ad564e45926	scikit-learn / joblib	—
M27	wing-walkers-in-proper-positi...	HRNet-W48 top-down heatmap pose (COCO-17)	mmpose TopdownPoseEstimator HRNet-W48 + HeatmapHead (same checkpoint ...	td-hm_hrnet-w48_8xb32-210e_coco-256x192-0e6...	269	0e67c6167d6a10fe	torch via mmpose 1.3.1	—

Shared and repeated: HRNet-W48 COCO 256×192 (sha 0e67c6167d6a10fe, 269 MB) is the same file in hand-signals, steering, lead-marshaller and wing-walkers, with the flip test on (two backbone passes per call); EfficientNet-B0 appears six times (the handrail classifier in two modules with the same weights, the difference classifier in aircraft-chocks and gse-chocks with different weights, the camera classifier in GM, two in hair policy); MobileSAM in four places (GM, tracker initialisation, safety-zone, the walk-around TDV build); five different pose stacks serve ten modules (HRNet, SimCC ResNet-50, RTMPose-I, DWPose-I, MoveNet, YOLOv8-pose). Present but loaded by nobody: HRNet-W48 384×288 (255 MB, in all four HRNet checkouts), wand_detector.pt, worker_type_classifier.pkl, v4_probe_head.npz, best28_full.onnx.

6 • Where the logic repeats, and what the review flagged

23 of 27 modules derive the turnaround stage themselves and 19 read the tracker's private optical-flow fields. The same arrival block (movement_anomaly_custom from _p0/_prev_p0/_st → normal_moving_counter > 24 → have_arrival_stage; pre-arrival = arrival_frame > 60-fps) is copied verbatim in nine modules; six more carry their own T_arr / T_dep rules — five definitions of departure in the code base. pushback_attached is computed twice (aircraft-chocks, pin-verification: IoU > 0.8 with the median pushback box of the first pass, nose-geometry fallback) and is the reason both need a second pass. “Belt loader at the door” exists in five variants (3-stop, beltloader-chocks and bl_rear_cone, handrails-on-gse, safety-handrails, hand-signals). The aircraft type is recomputed from wing/engine geometry in two modules although GM publishes it. Every frame read letterboxes to 1280×768 and copies the frame for nothing (cv_common datasets.py). Three modules run their own trackers beside the shared one (safety-zone, huddle, seat-belts).

M	module	events the module recomputes itself (file:line in the checkout)	private tracker fields read
M01	3-stop-brake-check	BL fully_stopped / re-init from _status counters (:76-102) · BL at door = get_relative_intersection(airplane.xyxy, bl.xyxy) > 0.2 (:190)	_status, _xyxy, _obj_id, _color
M02A	aircraft-chocks	arrival stage block: movement_anomaly_custom from _p0/_prev_p0/_st -> normal_moving_counter_custom>24 -> have... · pushback_attached: IoU>0.8 with median_pushback from pass 1 for >5*FPS frames, or nose-geometry fallback (:15... · BL in safety zone / removal frame from bl_type!= 'undefined' + rear-wheel geometry, in/out thresholds 16/40 (:...	_p0, _st, _of_dots_lifetime, _init_dots_lifetime, _obj_id (raw key)
M03	all-cargo-bin-doors-opened-and-verified	arrival stage block: movement_anomaly_custom from _p0/_prev_p0/_st -> normal_moving_counter_custom>24 -> have... · local stop_frame from status==STOPPED (:146-148) · local depart_frame from 5*fps MOVING frames (:149-152)	_p0, _st, _of_dots_lifetime, _init_dots_lifetime
M04	beltloader-chocks	BL at door = bl_type==needed + stopped (:234,:490) · aircraft type jet/airplane from wing/engine y2 (:326-368, min_plane_type_frames_eval 40)	_status (raw key), _obj_id, _color, _xyxy (written in dead code :288)

M	module	events the module recomputes itself (file:line in the checkout)	private tracker fields read
M05	bl_rear_cone	BL at door = bl_type==needed + stopped (:264,:446) · BL view side / enface from back_door geometry (:100-127)	_status (raw key), _obj_id, _color
M06	chocks-and-cones-available-and-staged-f...	arrival stage block: movement_anomaly_custom from _p0/_prev_p0/_st -> normal_moving_counter_custom>24 -> have... · HAD_ARRIVAL from 5 s of moving (:172-176, unused in verdict)	_p0, _st, _of_dots_lifetime, _init_dots_lifetime, _status
M09	cones-placed-in-proper-positions-and-ti...	T_arr = first frame with arrived (:62-63,:245-247) · T_dep = 30*fps MOVING frames after arrival (:252-258)	_status
M10	crew-present-10-minutes-prior-to-aircra...	arrival stage block: movement_anomaly_custom from _p0/_prev_p0/_st -> normal_moving_counter_custom>24 -> have...	_p0, _st, _of_dots_lifetime, _init_dots_lifetime
M11	fod-walk-completed	arrival stage block: movement_anomaly_custom from _p0/_prev_p0/_st -> normal_moving_counter_custom>24 -> have...	_p0, _st, _of_dots_lifetime, _init_dots_lifetime
M12	gse-chocks	GSE stop/departure moments from _static_frames==1 and _moving_frames==_display_proceeding_time_thres+1 (:1327... · no airplane anchors used at all	_xyxy, _p0, _st, _status, _static_frames
M13	hand-signals	plane_stopped from raw arrival_frame <= frame_number (:771-777) · BL approach/stop from _status + workers ROI (:461-520) · aircraft type jet/airplane (:675-717)	_status (raw, compared to enum .value), _obj_id, _color, _xyxy (dead code :627)
M14	handrails-on-gse-being-used	BL active = bl_type==needed + is_stopped (:726,:764)	_obj_id, _color
M15	lead-marshaller-and-wing-walkers-in-pos...	arrival stage block: movement_anomaly_custom from _p0/_prev_p0/_st -> normal_moving_counter_custom>24 -> have... · start_moving from first STOPPING/MOVING frame (:434-438) · departed_frame from 30*fps MOVING after arrival (:452-458, unused)	_p0, _st, _of_dots_lifetime, _init_dots_lifetime
M16	pin-verification	pushback_attached: pushback_frames > 10*fps with IoU>0.8 to median_pushback (pass 1) or nose geometry (:663-6...	_status
M17	post-arrival-aircraft-walk-around-inspe...	T_arr = first status==STOPPED (:397-399) · T_dep = 3*fps MOVING/STOPPING after stop (:422-426) · BL entering/leaving area from own velocity direction (:236-246,:543-555)	—
M18	pre-arrival-safety-huddle	arrival stage block: movement_anomaly_custom from _p0/_prev_p0/_st -> normal_moving_counter_custom>24 -> have...	_p0, _st, _of_dots_lifetime, _init_dots_lifetime
M19	pre-departure-walk-around-completed	T_arr = first status==STOPPED (:261-263) · T_dep = status==MOVING after stop, immediate (:264-267) · BL entering/leaving from own velocity direction (:352-369)	—
M20	pushback-does-not-start-until-wing-walk...	T_arr = STOPPED for fps*60 frames (:498-500) · T_dep = MOVING after stop (moving_frames, start_moving :502-505) · pushback attached = wheel-pushback distance < 315 px (:460, display only)	—
M21	pushback-pathway-confirmed-clear-of-obs...	pushback_moving = airplane status in [PROCEEDING, MOVING] (:370) · final_frame = departure_frame + 15*8 (:81,:187)	—
M22	safety-handrails-fully-extended	BL active = bl_type==needed + is_stopped + rear-wheel filter (:235-239,:269) · BL identity swap across track ids (:251-258)	_obj_id (raw key, rewritten), _color
M24	safety-zone-confirmed-clear	arrival stage block: movement_anomaly_custom from _p0/_prev_p0/_st -> normal_moving_counter_custom>24 -> have... · own norfair truck tracker + cv_common Vehicle instances with pixel optical flow (:990-994,:1420-1426)	_p0, _st, _of_dots_lifetime, _init_dots_lifetime
M26	steering-by-pass-pin-installed-or-steer...	arrival stage block: movement_anomaly_custom from _p0/_prev_p0/_st -> normal_moving_counter_custom>24 -> have...	_p0, _st, _of_dots_lifetime, _init_dots_lifetime
M27	wing-walkers-in-proper-position-and-usi...	T_arr = STOPPED for stop_thd*fps frames (:556-561) · T_dep = MOVING for depart_thd*fps frames with reset on STOPPED (:563-575) · fallback T_dep from statuses_airplane history at end (:772-795)	_status

The review's attention notes (the static audit of 7 September): what the active code verifies against what the checklist names.

M	module	attention
M01	3-stop-brake-check	Do not replace its recomputed status with generic tracker status without replay comparison; seconds vary with configured FPS.
M02A	aircraft-chocks	Placement is assessed within 30 seconds of aircraft stop. Pushback attachment and rear-wheel image differences are not prerequisites for this task.
M03	all-cargo-bin-doors-opened-and-verified	Does not verify a BL target-door relationship or independently distinguish mechanical open/closed state beyond detector-label semantics.
M04	beltloader-chocks	A one-time chock latch does not prove continuous chocking throughout the entire parked interval.
M05	bl_rear_cone	Checklist and code thresholds differ; preserve code behavior as the migration baseline, then approve requirement changes separately.
M06	chocks-and-cones-available-and-staged-for-a...	YAML 6/4 counts are not the active main-code 7/6 counts. Do not assume every configuration value controls behavior.
M07	conditioned-air-removed-10-mins-prior-to-de...	Held boxes are counted repeatedly in current history. Evidence is a visual area/position proxy, not an observed disconnected coupling.
M08	cones-are-removed-only-after-all-gse-is-cle...	Despite its name it does NOT test whether GSE is chocked, and its equipment coverage is narrower than all GSE. A chocked prerequisite is new functional work.
M09	cones-placed-in-proper-positions-and-timely	Obstacle report text says 40% while code tests 65%; inconsistent return arity requires an explicit legacy adapter.
M10	crew-present-10-minutes-prior-to-aircraft-a...	This measures counts over a window, not continuous presence of the same three named workers for ten minutes.
M11	fod-walk-completed	Not the simple horizontal-crossing rule in the basic description; bbox-derived scale is not calibrated physical distance.
M12	gse-chocks	Different chock evidence channels have different semantics; current truthy fusion must be made explicit before sharing.

M	module	attention
M13	hand-signals	Raised wrists are a coarse gesture proxy, not an approved signal vocabulary. Same HRNet checkpoint as steering does not imply same crop preprocessing.
M14	handrails-on-gse-being-used	Whole-body hands are required; do not substitute body-only pose. Unknown climbers are excluded rather than forcing global unknown in current aggregation.
M15	lead-marshaller-and-wing-walkers-in-position	No separate pose/wand model or lead-marshaller signal proof on the active path.
M16	pin-verification	Measures a verification-like worker action, not direct pin visibility. Current 1.5/2.5-second criteria differ from the supplied description.
M17	post-arrival-aircraft-walk-around-inspectio...	Post-arrival rescue/outlier options are disabled by default. Score-derived confidence is not demonstrated calibrated probability; required cached geometry must exist.
M18	pre-arrival-safety-huddle	Pinned cv_common ac5098d lacks imported PlaneArrivalDetector. A candidate implementation exists at cv_common 2759daf, not at the reviewed pin.
M19	pre-departure-walk-around-completed	Pinned cv_common e4c5e36 lacks imported departure/presence helpers. The reported five-minute wording does not match the 480-second code window. Torch and Keras paths must remain separate profiles.
M20	pushback-does-not-start-until-wing-walkers-...	Current evidence is post-start co-motion, not a direct proof that everyone was ready before start. Global role flags require per-turn state isolation.
M21	pushback-pathway-confirmed-clear-of-obstacl...	Current hazard class list does not include people. Variable pushback_moving uses aircraft state, not an independent pushback motion state.
M22	safety-handrails-fully-extended	480 frames is about 60 seconds at 8 FPS, not eight minutes. Shared classifier reuse needs artifact/preprocessing identity.
M23	safety-vests-secured-to-body	No keypoint pose despite the model name. Current unknown/undefined workers can coexist with Pass; do not silently strengthen semantics during extraction.
M24	safety-zone-confirmed-clear	Upstream tracker does not cover every transport class currently tracked here. FOD and safety-zone ROIs are related but not identical.
M26	steering-by-pass-pin-installed-or-steering-...	Distinct learned installation task from heuristic pin verification. Chock detections are collected but do not establish chocking prerequisite.
M27	wing-walkers-in-proper-position-and-using-a...	CRITICAL semantic gap: approved-wand Pass is hard-coded on the active path. Old wand-count code is not called; merely extracting it would not preserve active behavior or establish correctness.